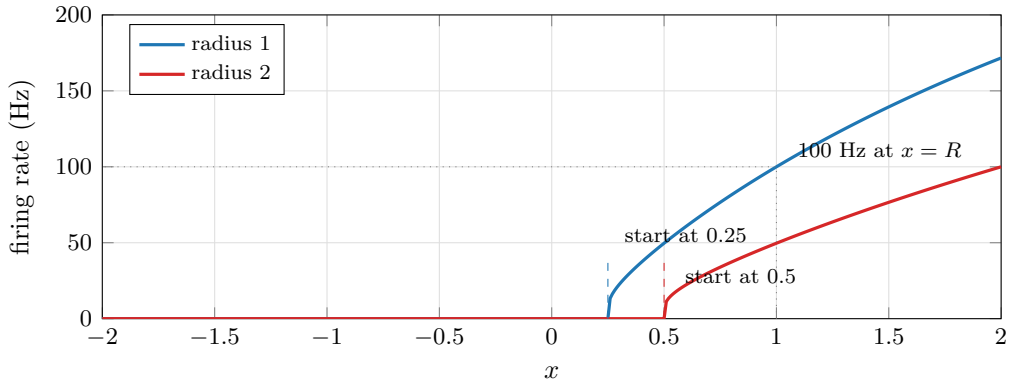
Practice Test 3C: Nengo & Simple Dynamics — Solutions

Total: 20 marks plus 2 bonus. Curves that come from a simulation use `pn3_data/make_data_3c.py`. Tuning curves use $\tau_{\text{ref}} = 2$ ms and $\tau_{RC} = 20$ ms. Any LIF shape with the labelled intercepts earns the mark.

Question 1 (4)

(a) [2] Nengo stores the intercept as a fraction of the radius, and defines the maximum rate as the rate at $\langle e, x \rangle = R$. Scaling the radius stretches the tuning curve by that same factor: $a_{2R}(x) = a_R(x/2)$.

- **Radius 1.** Firing starts at $x = 0.25$. The rate is 100 Hz at $x = 1$. Past the radius the current keeps growing, so the rate keeps rising (about 170 Hz at $x = 2$ with the time constants above). It does not fall back to zero.
- **Radius 2.** The same normalized intercept is now at $x = 0.5$, and 100 Hz is reached at $x = 2$. The curve is the radius-1 curve stretched horizontally by 2.



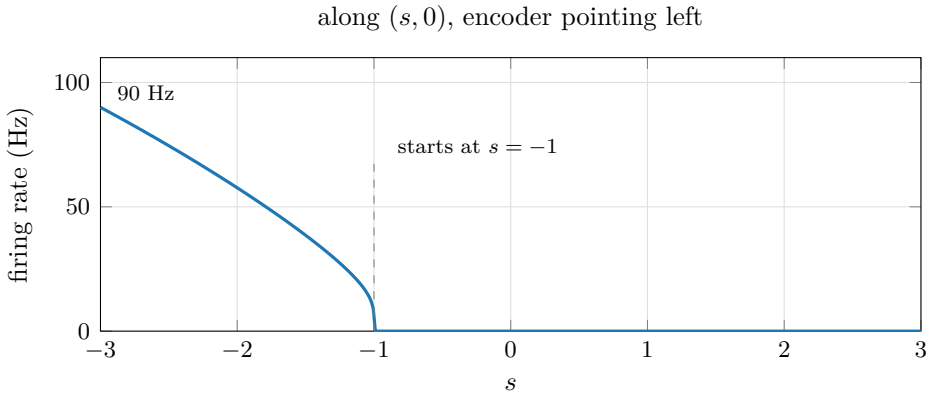
Marking

0.5: radius 1 starts at 0.25 and is at 100 Hz at $x = 1$. 0.5: radius 2 starts at 0.5 and is at 100 Hz at $x = 2$. 0.5: the radius-2 curve is the wider one. 0.5: past its 100 Hz point the radius-1 curve is still high.

Where marks are usually lost

Drawing the larger radius as a steeper curve, or dropping the rate to zero once x passes the radius. A bigger radius spreads the same curve over a wider interval.

(b) [2] The neuron sees $\langle e, \mathbf{x} \rangle / R$. Along $\mathbf{x} = (s, 0)$ with $e = (-1, 0)$ and $R = 3$, that is $-s/3$. It starts firing when $-s/3 = 1/3$, i.e. at $s = -1$, and it reaches 90 Hz when $-s/3 = 1$, i.e. at $s = -3$. For every $s > -1$ it is silent. The curve rises as s decreases.



Marking

1: zero to the right of $s = -1$, and firing starts at $s = -1$. 1: the rate grows as s decreases and reaches 90 Hz at $s = -3$.

Where marks are usually lost

A tuning curve that rises to the right. This encoder points in the negative direction, so the neuron prefers negative s.

Question 2 (4)

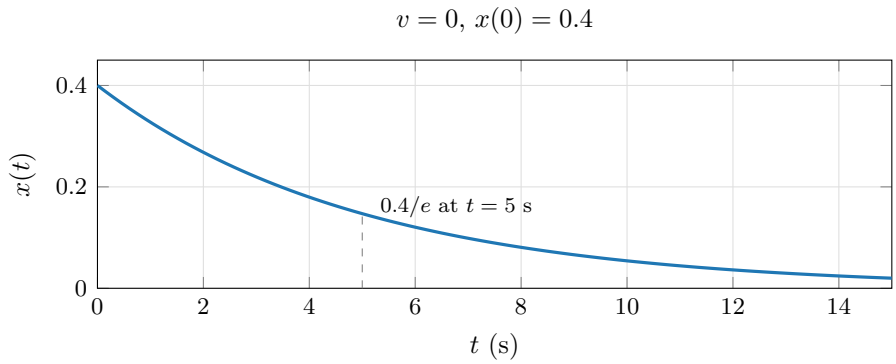
(a) [2] The target is $\dot{x} = Ax + Bv$ with $A = -1/5$ and $B = 1$. With $\tau = 0.1$,

$$A' = \tau A + 1 = 1 - \frac{0.1}{5} = \mathbf{0.98}, \qquad B' = \tau B = \mathbf{0.1}.$$

With $v = 0$ the system is $\dot{x} = -x/5$, so

$$x(t) = 0.4\, e^{-t/5}, \qquad x(5) = 0.4/e \approx \mathbf{0.147}.$$

The effective time constant is $\tau/(1 - A') = 0.1/0.02 = 5$ s, which is the 5 in the target system. The synapse is 0.1 s; the feedback cancels most of its leak and leaves a 5 s memory.



Marking

0.5: $A' = 0.98$. 0.5: $B' = 0.1$. 0.5: $x(t) = 0.4e^{-t/5}$. 0.5: $x(5) = 0.4/e$ (a value from 0.14 to 0.15 is fine).

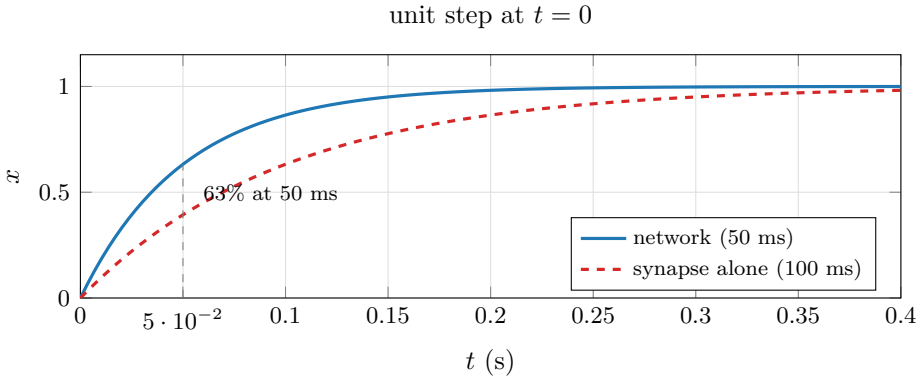
(b) [2] Now $A = -20$ and $B = 20$:

$$A' = 0.1(-20) + 1 = \mathbf{-1}, \qquad B' = 0.1(20) = \mathbf{2}.$$

A unit step settles at $x = u = 1$. The network time constant is $\tau/(1 - A') = 0.1/2 = \mathbf{0.05\ s}$, so

$$x(t) = 1 - e^{-20t}.$$

The synapse alone is $1 - e^{-t/0.1} = 1 - e^{-10t}$, which is slower. Negative feedback adds to the leak instead of cancelling it, and the large input transform $B' = 2$ keeps the steady state at 1. The loop can be faster than the synapse it is built from.



Marking

0.5 each: $A' = -1$, $B' = 2$. 0.5: both curves go to 1, and the network reaches it sooner (time constant 50 ms). 0.5: the network is faster because $A' < 1$ strengthens the leak ($A' = -1$ here).

Where marks are usually lost

Putting the 50 ms on the synapse. The synapse stays at 0.1 s. Also, $A' = \tau A$ without the $+1$ gives -2 instead of -1.

Question 3 (4)

(a) [2]

$$\tau\dot{x} = 0 + x^3 - x = \boxed{x^3 - x} = x(x-1)(x+1).$$

Equilibria at $x = -1,\ 0,\ 1$. The slope is $\varphi'(x) = 3x^2 - 1$:

- $x = 0$: $\varphi' = -1 < 0$, **stable**. Between -1 and 1, φ pushes back toward 0.
- $x = \pm 1$: $\varphi' = 2 > 0$, **unstable**. Outside that interval φ has the same sign as x, so the state runs away.

Marking

1: $\varphi(x) = x^3 - x$, with the synapse's $-x$ included. 0.5: all three equilibria. 0.5: 0 stable and ± 1 unstable, with a reason.

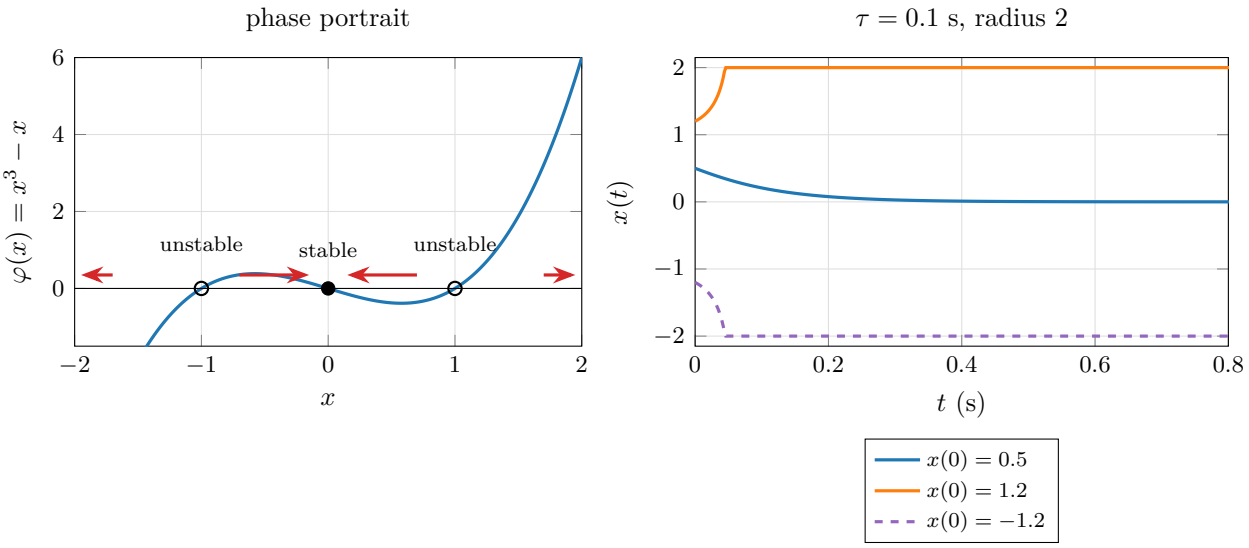
Where marks are usually lost

Dropping the $-x$, which leaves only the equilibrium at 0 and makes it unstable. The cubic $x - x^3$ has the opposite stability: there the outer points are the stable ones.

(b) [2]

- $x(0) = 0.5$ is between the unstable points, and $\varphi(0.5) < 0$, so x decreases and settles at **0**.
- $x(0) = 1.2$ is past the unstable point at 1. $\varphi(1.2) > 0$, so x increases until the population saturates at the radius, $+$**2**.

The trajectories below use $\tau = 0.1$ s and stop at the radius.



The third curve is the mirror of the second: $x(0) = -1.2$ saturates at -2. It is not asked, but it is the same rule.

Marking

0.5: sketch with $\varphi(0) = 0$ stable and crossings at ± 1 unstable, arrows inward between them and outward outside. 0.5: $x(0) = 0.5$ goes to 0. 1: $x(0) = 1.2$ runs away to the positive saturation.

Question 4 (4)

(a) [1] $\tau\omega = 0.05\pi = \pi/20$:

$$A' = \begin{bmatrix} 1 & -\pi/20 \\ \pi/20 & 1 \end{bmatrix} \approx \begin{bmatrix} 1 & -0.157 \\ 0.157 & 1 \end{bmatrix}.$$

Marking

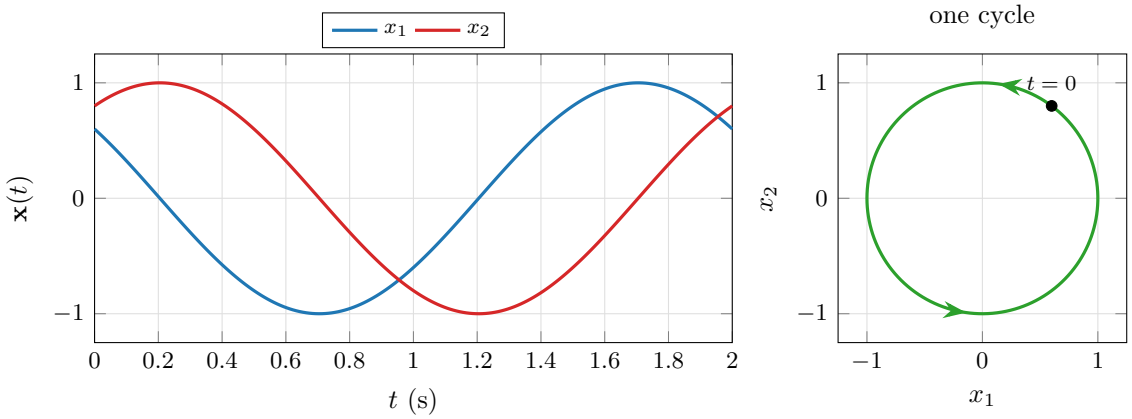
0.5: ones on the diagonal. 0.5: off-diagonal entries $\mp\pi/20$ with the same sign pattern as A.

(b) [3] Rotating the initial state by the angle $\omega t = \pi t$ gives

$$x_1(t) = 0.6\cos(\pi t) - 0.8\sin(\pi t),$$
$$x_2(t) = 0.6\sin(\pi t) + 0.8\cos(\pi t).$$

Since $0.6^2 + 0.8^2 = 1$, both signals have amplitude 1. The period is $2\pi/\omega = \mathbf{2}$ **s**, so the window is one cycle. Equivalently $x_1 = \cos(\pi t + \phi)$ and $x_2 = \sin(\pi t + \phi)$ with $\cos\phi = 0.6$ and $\sin\phi = 0.8$.

At $t = 0$ the state is $(0.6, 0.8)$. There $\dot{x}_1 = -\omega x_2 < 0$ and $\dot{x}_2 = \omega x_1 > 0$, so it moves left and up: **counter-clockwise** around the unit circle.



Four easy checkpoints, a quarter cycle apart: $(0.6, 0.8)$ at $t = 0$, $(-0.8, 0.6)$ at $t = 0.5$, $(-0.6, -0.8)$ at $t = 1$, $(0.8, -0.6)$ at $t = 1.5$.

Marking

1: both formulas, or the equivalent shifted sine and cosine of amplitude 1. 0.5: period 2 s. 0.5: the time plot starts at 0.6 and 0.8, with x_1 decreasing. 0.5: circle of radius 1. 0.5: counter-clockwise.

Question 5 (4)

(a) [2]

$$x(t) = \begin{cases} 3t, & 0 \le t < 0.4, \\ 1.2, & t \ge 0.4. \end{cases}$$

A straight ramp of slope 3, ending at 1.2, then a flat hold. The corner is at $t = 0.4$ s.

Marking

1: $3t$ on $[0, 0.4)$. 0.5: holds at 1.2. 0.5: the sketch is a straight ramp, then flat.

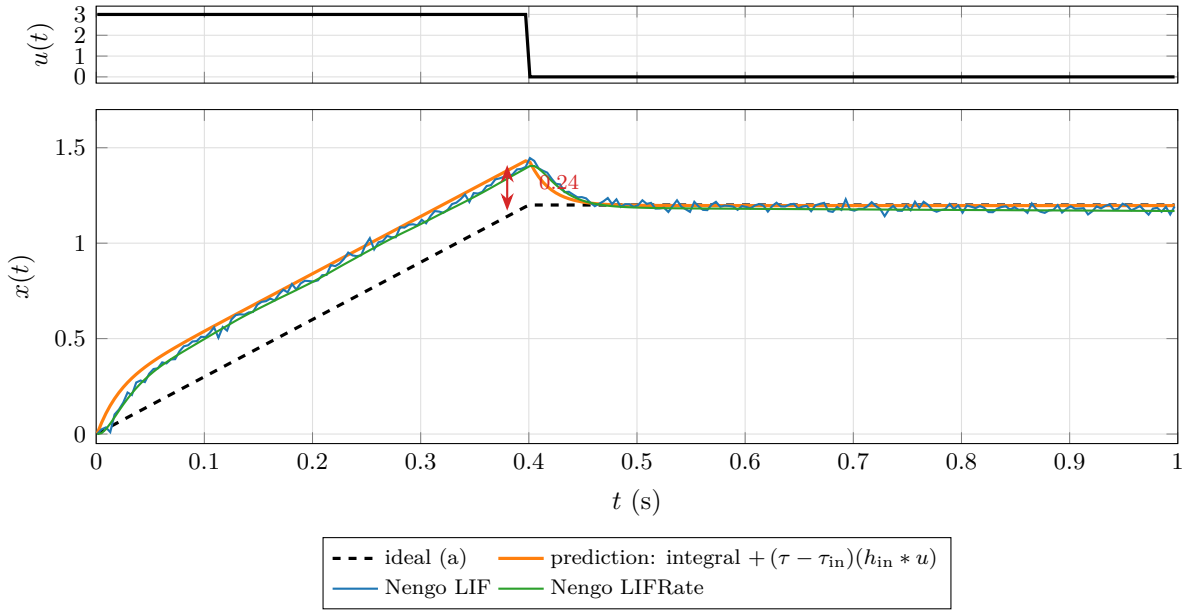
(b) [2] The two synapses disagree, so

$$X = \frac{U}{s} + (\tau - \tau_{\text{in}})\frac{U}{1 + s\tau_{\text{in}}}.$$

The second term is $(\tau - \tau_{\text{in}})$ times the input smoothed by the fast synapse. While the pulse is on, $u = 3$ and $\tau_{\text{in}} = 20$ ms is short, so the gap is

$$(\tau - \tau_{\text{in}})\, u = (0.1 - 0.02) \cdot 3 = \mathbf{0.24}.$$

Just before $t = 0.4$ s the decoded value is about $1.2 + 0.24 = 1.44$, above the ideal ramp. When the pulse ends, $u = 0$ and the extra term dies within a few τ_{in}. The stored integral is untouched, so x settles at **1.2**.



The spiking trace sits slightly under the prediction because the readout probe adds its own filter. The final value is still 1.2.

Marking

1: the trace lies a constant gap above the ideal ramp while the pulse is on. 0.5: the gap is 0.24 (accept 0.2 to 0.25). 0.5: after the pulse it returns to 1.2.

Where marks are usually lost

Reading the faster input synapse as a small delay. The error is an offset of size $(\tau - \tau_{\text{in}})u = 0.24$, and it disappears once u returns to 0.

Bonus Question 1 (2)

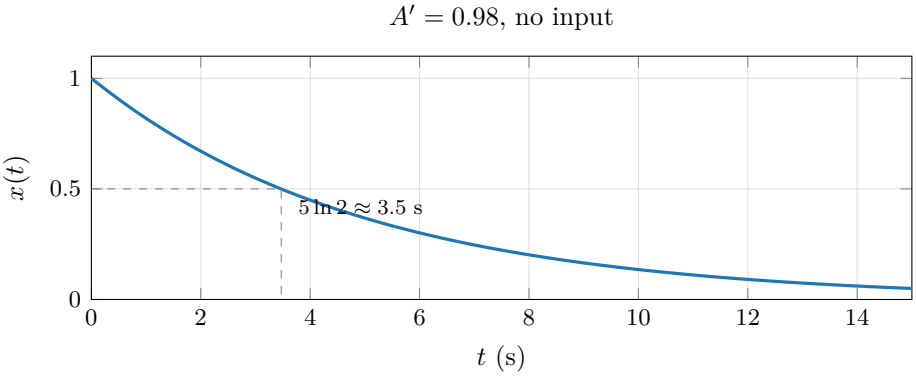
Recurrent transform $A' = 0.98$ with $\tau = 0.1$ s and no input means

$$A = \frac{A' - 1}{\tau} = \frac{-0.02}{0.1} = -0.2, \qquad \dot{x} = -0.2\,x.$$

The time constant is $\tau/(1-A') = 0.1/0.02 = 5$ s, so $x(t) = e^{-t/5}$. It falls from 1 to 0.5 when $e^{-t/5} = 1/2$:

$$t = 5 \ln 2 \approx \mathbf{3.47}\text{ s}.$$

A recurrent gain 2% below 1 turns a perfect memory into a 5 s leak. The same 2% above 1 would make the held value grow until it hit the radius.



Marking

1: $\dot{x} = -0.2x$, or a leak with time constant 5 s. 1: about 3.5 s to fall from 1 to 0.5, with a decaying sketch.